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School of Computer Science Engineering and Information Systems

Department of Information Technology

BITE497J • Project – I • Review 1

CogniShield

Edge-Based Adversarial Immunization for Digital Media

Presented by

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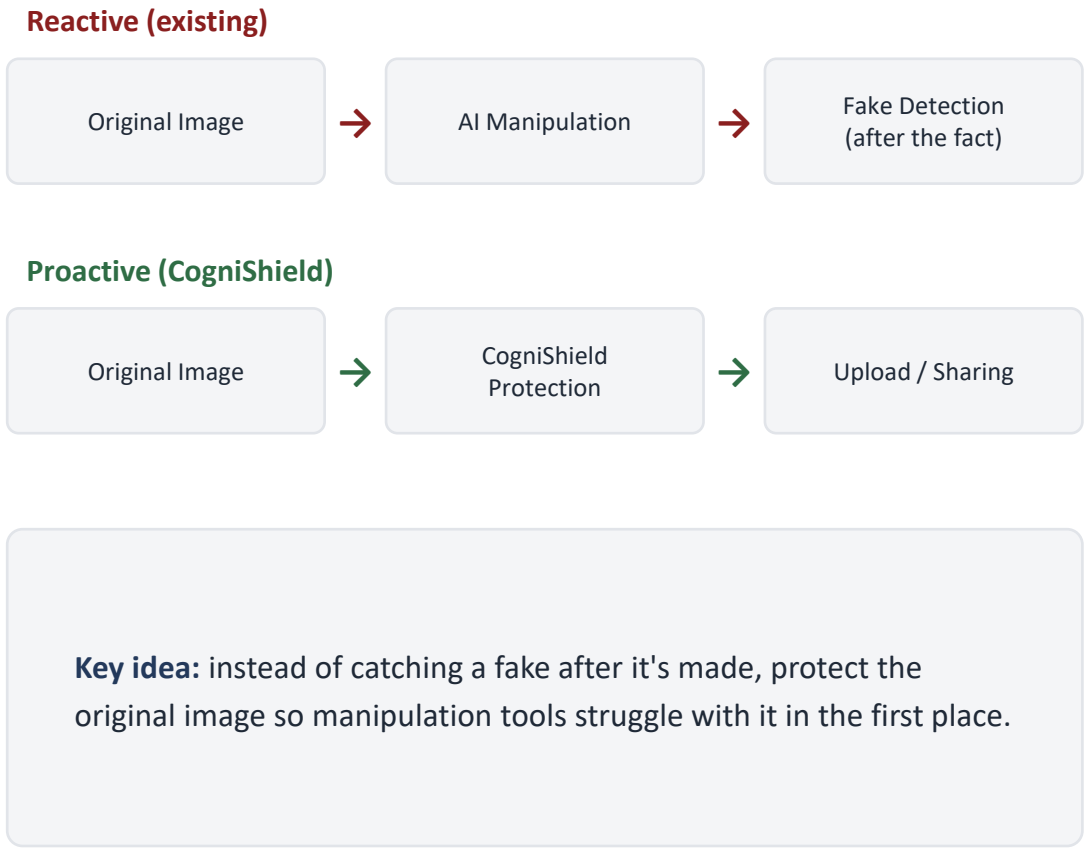
Under the guidance of

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Fall Semester 2026–2027

Problem Statement

- Generative AI has made image editing and face manipulation very easy.
- Personal photos can be misused to create fake or manipulated images.
- Most existing systems only detect a deepfake after it has already been created.
- By the time detection happens, the manipulated image may already be shared online — the damage is done.



Key idea: instead of catching a fake after it's made, protect the original image so manipulation tools struggle with it in the first place.

Existing Approach vs Proposed Approach

REACTIVE

Existing Approach

- Detects a fake image after it has already been generated.
- Relies on classifiers trained to spot manipulation artifacts.
- Detectors often don't generalize well to new/unseen generative models.
- By the time it's flagged, the image may already be circulating.

PROACTIVE

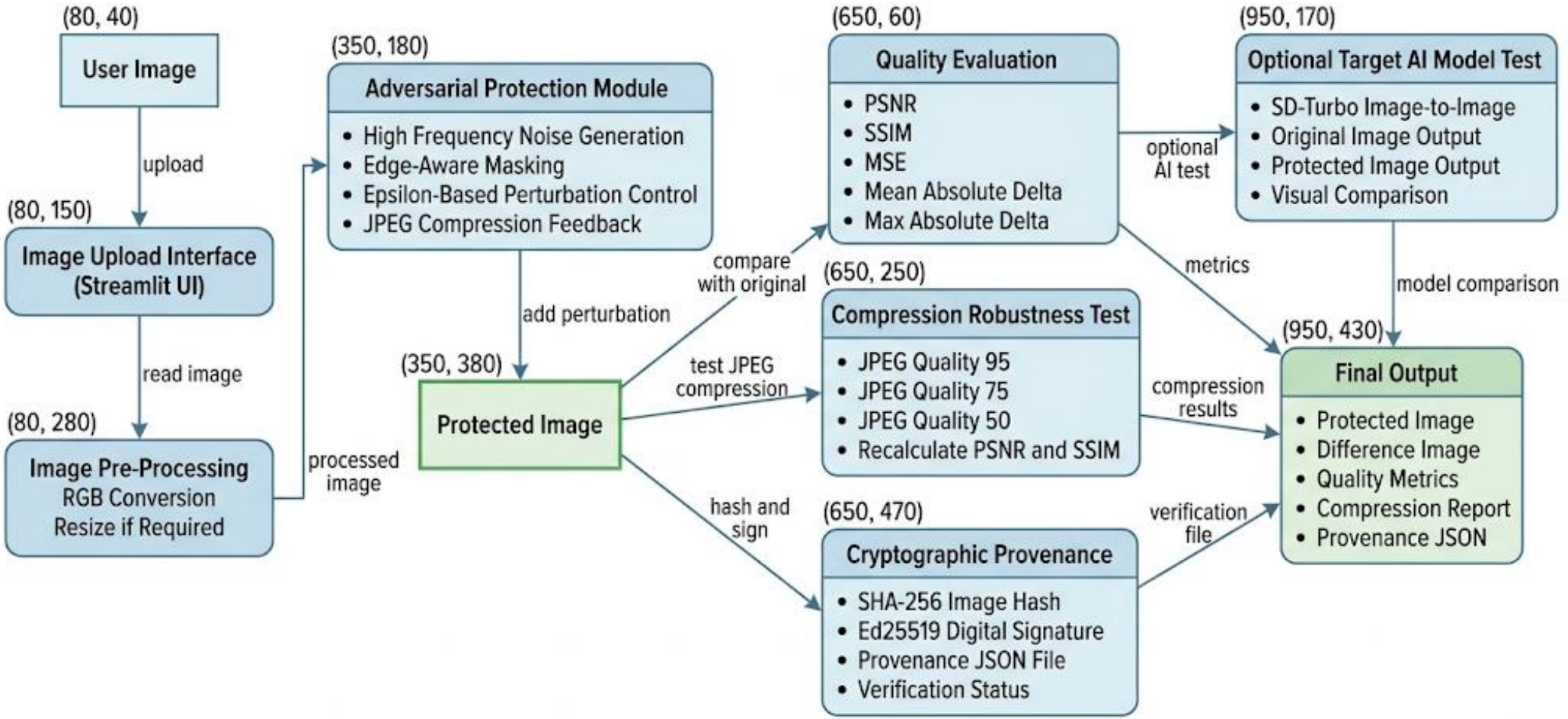
Proposed Approach (CogniShield)

- Protects the image before it is uploaded or shared.
- Adds low-visibility perturbations that interfere with AI manipulation models.
- Runs locally on the device — no upload to a third-party server.
- Adds provenance so tampering can be checked later.

Objectives

1. Low-visibility image protection using adversarial-style perturbation
2. Reduce effectiveness of AI-based image manipulation models
3. Preserve visual quality of the protected image
4. Test robustness of protection against JPEG compression
5. Add cryptographic provenance for tamper verification
6. Run entirely locally / at the edge for privacy
7. Provide a simple user interface (Streamlit)

System Overview / Architecture



How the Protection Works

High-frequency perturbation

Small noise is added that is difficult for the human eye to notice, but disturbs how AI models read the image.

Edge-aware mask

The perturbation is guided to blend into natural edges and textures rather than flat, smooth areas.

Epsilon / perturbation budget

A fixed limit (epsilon) controls the maximum pixel-level change, so the image stays visually close to the original.

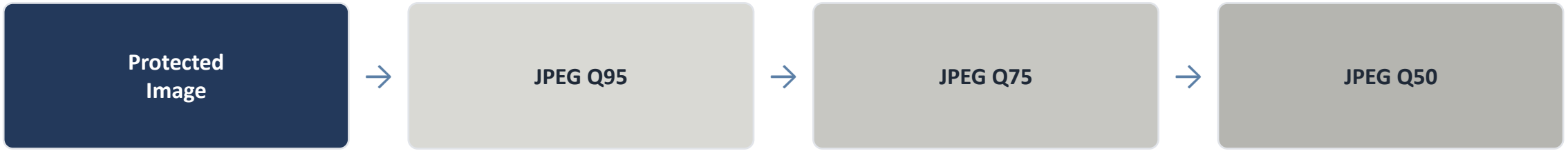
JPEG compression feedback

Compression is simulated during protection, since platforms often compress images after upload.

Image Quality Evaluation

Metric	What it measures	Direction
PSNR	Peak Signal-to-Noise Ratio	▲ Higher is better
SSIM	Structural Similarity Index	▲ Higher is better
MSE	Mean Squared Error	▼ Lower is better
Mean Abs. Pixel Difference	Average per-pixel change	▼ Lower is better
Max Pixel Difference	Largest single-pixel change	▼ Lower is better

JPEG Compression Robustness



Quality decreases left → right (95 = light compression, 50 = heavy compression)

Why compression matters

- Social media and messaging platforms almost always re-compress uploaded images.
- Compression can smooth out or discard the tiny pixel-level changes that make up the protection.
- So a protection method has to be checked under compression, not just on the raw file.

What CogniShield does

- Applies compression feedback during protection itself.
- Then re-tests the protected image at JPEG quality 95, 75, and 50.
- Recomputes PSNR / SSIM after each compression level to see how much protection survives.

Cryptographic Provenance



↑ *Verification checks the hash against the stored signature*

What each part does

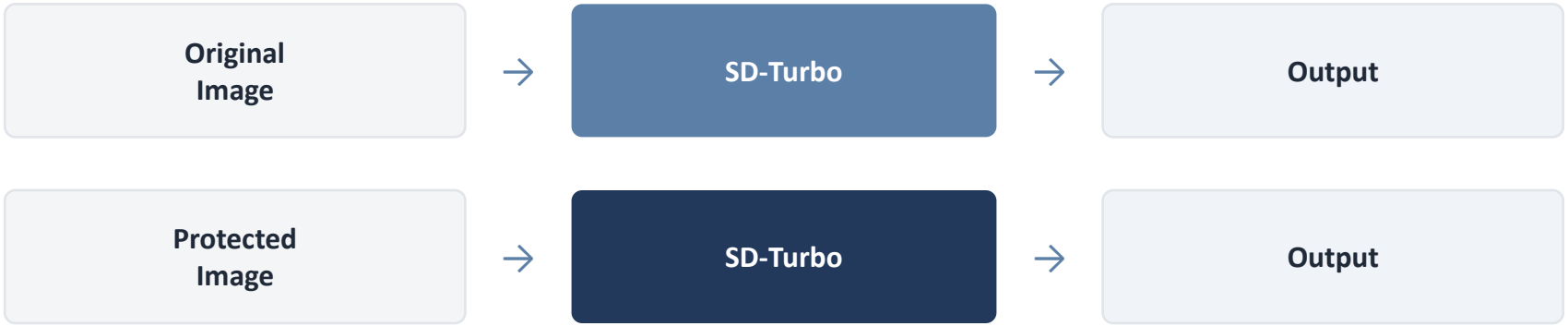
SHA-256: creates a unique fingerprint of the protected image.

Ed25519: digitally signs that fingerprint using a private key, so authenticity can be verified later with the public key.

Important

Provenance does NOT prevent a deepfake from being created. It only helps verify whether a signed, protected file has been modified after it was signed — it is a complementary integrity check, not a protection mechanism on its own.

Optional SD-Turbo Testing



Compare the two outputs visually

What we're checking: does the AI model produce a visibly different / degraded result on the protected image compared to the original, when given the same prompt and settings?

Note: SD-Turbo testing is an optional evaluation component to observe target-model behavior — it is not the core protection algorithm.

Technology Stack

Image Processing

Pillow, OpenCV, NumPy

User Interface

Streamlit

Quality Metrics

scikit-image

Security / Provenance

SHA-256, cryptography, Ed25519

AI / Target-Model Testing

PyTorch, Hugging Face Diffusers, SD-Turbo

Language

Python

Literature Survey / Research Gap

Adversarial Image Immunization

DiffusionGuard, Distraction is All You Need, Glaze

Compression / Robustness Defenses

DCT-Shield, JPEG-Compression Study

Deepfake Detection

Detector Generalization Studies (Li et al., UCF)

Provenance / Authentication

C2PA Security Analysis

Research Gap

Prior work studies adversarial protection, compression robustness, deepfake detection, and provenance mostly as separate components. CogniShield's research gap is combining proactive protection, measurable quality evaluation, compression-robustness testing, local/edge computation, and complementary cryptographic provenance into one lightweight pipeline.

Limitations & Future Work

Current Limitations

- Perturbation method is prototype-level, not full adversarial optimization.
- Not guaranteed to work against every generative AI model.
- Cropping, resizing, denoising, purification, and heavy compression can weaken protection.
- Current system covers images only — no video or audio yet.

Future Work

- Full gradient-based adversarial optimization (e.g. PGD).
- Testing against more target generative models.
- Mobile / edge deployment using ONNX or TensorFlow Lite.
- Broader testing on real social-media upload/download pipelines.

Conclusion

- CogniShield provides a proactive approach to protect digital images from generative AI misuse.
- Uses adversarial perturbations to reduce the effectiveness of AI-based image manipulation.
- Evaluates image quality using PSNR, SSIM, MSE, and pixel-difference metrics.
- Tests protection robustness against JPEG compression.
- Uses SHA-256 and Ed25519 for image integrity and cryptographic provenance.
- Runs locally, providing privacy-preserving edge-based processing.
- The current prototype establishes a foundation for stronger AI-image protection and future mobile deployment.

THANK YOU